

- 6 (a) (i) period = $1/50$
 $t_1 = 0.03 \text{ s}$ C1
A1 [2]
- (ii) peak voltage = 17.0 V A1 [1]
- (iii) r.m.s. voltage = $17.0/\sqrt{2}$
= 12.0 V A1 [1]
- (iv) mean voltage = 0 A1 [1]
- (b) power = V^2/R C1
= $12^2/2.4$
= 60 W A1 [2]
- 7 (a) each line represents photon of specific energy M1